RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic year 2025-2026 (Odd Sem)

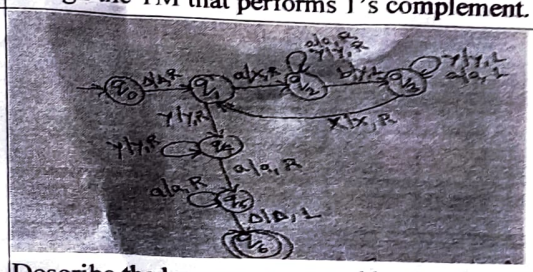
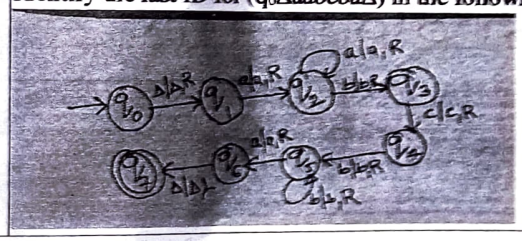
DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

Date	13-01-2026	Maximum Marks	10+50
Course Code	CS354TA	Duration	120 Min
Sem-V	Improvement Test	Staff: HKK|ASP|SMS|CRM|DL|SBP|MRA|PK	

THEORY OF COMPUTATION
(Common to CSE & ISE)

PART-A

		Marks	BT	CO
1.	Design the TM that performs 1's complement.	2	L3	CO3
2.	 Describe the language accepted by the above TM.	2	L2	CO2
3.	Write a Linear Bounded Automaton that accepts the language L={w ∈ {0,1}* | w contains 001 as a substring}	2	L3	CO3
4.	Write CSG to generate the language L = { $a^n b^m c^n$: n, m>0}. Give the derivation for the string aabbcc.	2	L3	CO3
5.	Identify the last ID for (q_0ΔaabcbaΔ) in the following Turing Machine. 	2	L3	CO4

PART-B

1.	Define Turing Machine, Language acceptance by Turing Machine. Construct Turing machine to accepting the language L={$a^i b^i c^i d^i$ | i≥0}. Using Instantaneous Descriptions show that the string aabbccdd is accepted by TM.	10	L4	CO4
2.	Design a Turing Machines (TM) to compute the function f(x, y)=xy where x, y ∈ {a,b,c}*. Trace the machine for x=bcab and y= abca.	10	L4	CO4
3a.	Define Linear Bounded Automata, Language acceptance by Linear Bounded Automata. Design Linear Bounded Automata to accept L= { w | w ∈ {a, b}* and $N_a(w)=N_b(w)$}.	6	L4	CO4
3b.	Explain the following with neat sketch diagram: i. Multi-tape Turing machine and Multitrack Turing machine ii. Restricted Turing machine and Turing machine with stay option	4	L2	CO2

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4a.	Define PCP. Solve the PCP given below. i: List A: wi; List B: Xi 1: 10; 101 2: 01; 100 3: 0; 10 4: 100; 0 5: 1; 010	5	L3	CO3
4b.	With the help of Chomsky hierarchy explain the relationship among families of languages.	5	L2	CO2
5a.	Define Unrestricted Grammars. Write Unrestricted grammar to generate language $L = \{ a^n b^n a^n b^n \}$. Give the derivation for the string aabbaabb.	6	L4	CO4
5b.	Prove that if L_1 and L_2 are both recursively enumerable languages over Σ, then $L_1 \cup L_2$ and $L_1 \cap L_2$ are also recursively enumerable.	4	L2	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	-	11	15	34	-	-	15	13	32	-	-

Course Outcomes:

1. Understand the fundamental concepts of theory of computations
2. Analyze the tools of finite automata to various fields of computer science.
3. Design solution model for complex problems, using the appropriate skills of automata theory for better results.
4. Apply automata skills in situations that describe computation effectively and efficiently.

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